Raspberry Pi Pico

P1
rpi pico 1
2648-SC0915CT-ND
tht

+3.3V
R1 220
3V1 green
GND

R10 220k
GND

+BATT
+3.3V
GND
J1
5V

NT5 NetTie_2
NT6 NetTie_2
NT7 NetTie_2
NT8 NetTie_2

bt1 bt2 bt3 bt4

1.8–5.5v
5V
D1 D_Schottky
+3.3V

VBUS VSYS 3V3_OUT ADC_VREF
3V3_EN RUN
GP0 GP1 GP2 GP3 GP4 GP5 GP6 GP7 GP8 GP9 GP10 GP11 GP12
GP13 GP14 GP15 GP16 GP17 GP18 GP19 GP20 GP21 GP22 GP26 GP27 GP28
AGND GND GND_1 GND_2 GND_3 GND_4 GND_5 GND_6

sda scl
right_1 right_2 right_en
left_2 left_en left_1

R2 220
D13 blue
GND

+3.3V
J3 joystick
GND

+3.3V
J4 speed knob
GND

left motor

U1
DRV8245HQPWPRQ1

LVL6, in independent mode that means faults are retries

SR DIAG PH/IN2 EN/IN1 DRVOFF VM VM VM OUT1 OUT1 OUT1 GND GND GND
left_2 left_1
+BATT GND
C1 0.1uF

EPAD ITRIP MODE NFAULT IPROPI NSLEEP VM VM VM OUT2 OUT2 OUT2 GND GND GND
EPAD GND GND
8.2k +-10% LVL2 INDEPENDENT
8.2k R7 GND
left_ipropi left_en
+BATT

HSL1 Heatsink

J100 0197054001
J101 0197054001
optionally replace J7 to be weelye compatible

ML+ ML−
J7 1017491

right motor

U2
DRV8245HQPWPRQ1

SR DIAG PH/IN2 EN/IN1 DRVOFF VM VM VM OUT1 OUT1 OUT1 GND GND GND
right_1 right_2
+BATT GND
C2 0.1uF

EPAD ITRIP MODE NFAULT IPROPI NSLEEP VM VM VM OUT2 OUT2 OUT2 GND GND GND
EPAD GND GND
8.2k +-10% LVL2 INDEPENDENT
8.2k R6 GND
right_ipropi right_en
+BATT

HSR1 Heatsink

MR− MR+
J8 1017491
optionally replace J8 to make the PCB weelye compatible
J104 2159151002

ADC (for measuring battery voltage and motor current)

U4
TLA2024IRUGR

+3.3V NT1 NetTie_3
GND TestPoint TP0
+BATT
R13 100k
R14 13.7k
GND

ADDR AIN0 AIN1 AIN2 AIN3 NC VDD SCL SDA GND
right_ipropi left_ipropi
scl sda
+3.3V
C4 0.1uF
GND

TP1 TestPoint left_ipropi
TP2 TestPoint right_ipropi
R11 604
R12 604
GND GND

+3.3V
R20 EXB34V182JV
pullup resistors
1.8k
sda scl

qwiic I2C plug
GND +3.3V sda scl
J2 Conn_01x04
MP

weelye compatible header
J11 Conn_01x10_Pin
+BATT switch
D2 D_Schottky
+BATT
bat−
GND

3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick
B1 B2 B3 B4
ASJ-200-B-HT
bt1 bt2 bt3 bt4
GND

battery power circuit
+BATT
J102 0197054001
J103 0197054001
optionally replace J5 to be weelye compatible
J5 TB005-762-02BE
bat−
R3 8.2k
18V zener D3
SW1 screw terminal
C5 33uf
R4 590k
switch
J6
PSMN1R2-25YLDX NMOS Q1 reverse polarity protection
PSMN1R2-25YLDX Q2 on/off switch
GND

3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm
H1 M3
H2 M3
H3 M3
H4 M3

alternative voltage regulator
Instead of the voltage regulator circuit, you could get a single THT module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input voltage are compatible with your battery.
U5
D78B05T, RBT05W24S05, pololu5593
+BATT IN GND OUT 5V
GND

voltage regulator circuit https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=F7F7D983B50C08E2
+BATT
R5 590k
U3
TPS54202HDDCR
EN FB VIN BOOT SW GND
C6 0.1uF
L1 12uH
C7 56pF
R8 100k
R9 13.7k
C8 47uF
C3 10uF
5V
GND

https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com
Sheet: /
File: gbg-pcb.kicad_sch
Title:
Size: A3 Date: Rev:
KiCad E.D.A. 10.0.1 Id: 1/1